Exercise Sheet

Instructor: Fabio Patrizi

December 16, 2025 - v.1

Table 1: Dataset

	x_1	x_2	y_1	y_2
0	1.0	5.0	0	2.6
1	2.0	6.0	0	3.4
2	3.0	5.5	0	3.5
3	4.0	6.5	0	3.7
4	5.0	5.0	0	6.2
5	6.0	4.5	0	6.6
6	7.0	3.5	0	4
7	6.5	2.5	0	1.8
8	3.5	3.0	1	2.2
9	4.5	3.5	1	3.5
10	5.0	4.0	1	4.6
11	4.0	4.0	1	4.4
12	3.0	4.0	1	5
13	2.5	3.5	1	1.5
14	5.5	3.0	1	2

Exercise 1 (KNN)**.**

- *Given the dataset obtained by taking only the components x_1, x_2, y_1 from Table 1, where $y_1 \in \{0, 1\}$ represents a class, use KNN to classify point $\mathbf{x} = (4, 5)$ (triangle), for $K = 1, 3, 5$. Describe the procedure and the formulas used.*
- *What is the computational complexity of classifying a point?*
- *In order to guarantee that KNN yields a classification error lower than Bayesian Optimal Classifier's, how should the size of the dataset grow when the data dimensionality d increases? (You don't have to detail the formal relationship between the two, a qualitative response is sufficient).*
- *Given the dataset obtained by taking only the components x_1, x_2, y_2 from the table above, where $y_2 \in \mathbb{R}$ represents the output, use KNN for regression on point $\mathbf{x} = (4, 5)$, for $K = 1, 3, 5$. Describe the procedure and the formulas used.*

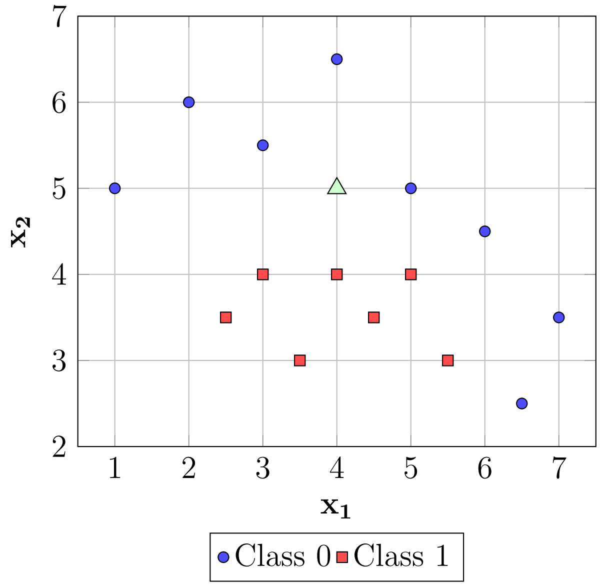

Figure 1: Dataset in $\mathbb{R}^2$

Exercise 2 (Clustering)**.**

- *Describe the application of Max-Linkage Clustering to the dataset obtained by taking only the components x_1, x_2 from Table 1. For convenience, Fig. 1 also shows the data in $\mathbb{R}^2$ (you can disregard colors an the triangle).*

- *Are there other forms of Linkage-Based clustering? How do they differ from Max-Linkage?*

- *Describe the application of the K-Means algorithm to the same dataset as that of the previous exercise (i.e., components x_1, x_2 only), for $K = 3$, starting with an arbitrary configuration.*

- *What are other variants of K-Means? How do they differ from each other?*

- *Is Clustering a formally defined problem? Discuss any related issue.*

Exercise 3 (Neural Networks)**.**

- *Design a FNN for multiclass classification, with 3 classes, having depth 3, with each hidden layer of width 4, and taking a 4-dimensional input. The i-th hidden layer must be represented by the corresponding matrix Ω_i and bias vector β_i. Report also a graphical representation of the network and indicate which activation functions are used in each layer.*

- *Comment on the following statements:*

  - *Stochastic Gradient Descent is not appropriate for training a Neural Network.*

- *Stochastic Gradient Descent is guaranteed to minimize the loss function associated with a Neural Network up to an arbitrary error, provided its parameters are suitably set.*
- *Backpropagation is the correct training algorithm for Neural Networks as it minimizes the gradient of the loss function.*

- *Define a CNN for regression, taking as input an RGB image (3 channels) of size* 10×10, *with two convolutional layers, both using a kernel of size 3 and zero padding. The first layer has stride 1 and outputs 2 channels; the second layer has stride 2, dilation 2, no padding, 'xand outputs one channel. Define the remaining part to return a unidimensional output.*
- *Explain the notion of* receptive field.
- *Explain the notion of Max Pooling and its purpose.*
- *What do we mean by* downsampling *and* upsampling*? When is the latter useful?*

Exercise 4 (Dimensionality Reduction)**.**

- *Explain the purpose of Principal Component Analysis (PCA) and Autoencoders in Machine Learning. Describe the two approaches, highlighting differences, similarities, and their relationship.*
- *Consider a dataset* $S = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ *with input data* $x_i \in \mathbb{R}^4$ *having mean* μ*, and let*

$$\Sigma = \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu)(x_i - \mu)^T = \begin{pmatrix} 2.0 & 0.8 & 0.6 & 0.4 \\ 0.8 & 1.5 & 0.5 & 0.3 \\ 0.6 & 0.5 & 1.2 & 0.2 \\ 0.4 & 0.3 & 0.2 & 0.8 \end{pmatrix}$$

be the respective covariance matrix. *Also, let*

$$Q = \begin{pmatrix} 0.64 & -0.48 & 0.47 & 0.38 \\ 0.53 & 0.75 & -0.23 & -0.31 \\ 0.43 & -0.09 & -0.78 & 0.44 \\ 0.35 & -0.45 & -0.35 & -0.74 \end{pmatrix}$$

be the matrix whose i-th column is the eigenvector of Σ *for the i-th largest eigenvalue* λ_i. *Define a linear transformation* W *of the normalized input data into* $\mathbb{R}^2$ *and an inverse transformation* U *s.t* U *and* W *minimize, together, the* reconstruction error *of the normalized data, i.e.,*

$$(U, W) = \arg\min_{U,W} \sum_{i=1}^{m} ||(x_i - \mu) - UW(x_i - \mu)||^2$$

- *How would the above procedure work on the original (non-centered) data?*

- *Define the architecture of an autoencoder with a total of 4 hidden layers and one bottleneck to solve the above problem. Once the autoencoder is trained, assuming $y_i \in \mathbb{R}$, how can it be used for regression?*

Exercise 5 (Ensemble Methods)**.**

- Report the AdaBoost algorithm and provide a qualitative description, explaining, in particular, the intuition behind the data distribution update step.
- Explain the notion of *Weak Learner*.
- What is the impact of an AdaBoost iteration on the training error?
- Describe the Bagging approach for regression.
- Explain the dataset sampling procedure for the Bagging approach. Do the bootstrapping sets partition the dataset?